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A review on the role of artificial intelligence and data science in green and sustainable chemistry: current trends and futuristic pathways

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Abstract
The combination of artificial intelligence, data science and green chemistry is leading scientific activity in addressing the planetary scale environmental challenges. This collective approach is to minimise waste and maximise energy efficiency, in addition to safer chemical design. Green chemistry is pivotal for developing more sustainable processes and production systems that minimize harm to human health and the environment. The integration of artificial intelligence and data science is transforming green chemistry. These tools can now analyze very large and complex datasets, predict chemical behaviour and help optimize processes to make them more sustainable. This review highlights recent advances in artificial intelligence for reaction optimization, new material discovery, process automation, and real-time environmental monitoring. It shows how machine learning, cheminformatics and data analytics work together to accelerate innovation in green and sustainable chemistry. Although this transformation is promising, major challenges still exist, such as the lack of high-quality data, the difficulty in interpreting AI models, and the limited collaboration across disciplines. To overcome these barriers, the development of data standards, open platforms, and interconnected research ecosystems is needed. By unlocking the potential of artificial intelligence and data science for green chemistry, this work highlights a strategic route to sustainable development goals achievement and to a cleaner more sustainable future.

Keywords Green chemistry, Sustainability, Artificial intelligence, Nanomaterial, Eco-friendly

1 Introduction

In recent decades, the evolution of societal and ethical framework has lagged significantly behind scientific and technological progress. This gap creates existential threats along with innovative possibilities. A key issue that emerges from this is how practices in academia and industry can be synchronized with sustainability principles. To solve

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the issue, the field of green chemistry emerged to prominence as an important scientific area in the 1990s. The goal is to redesign chemical processes and products to reduce their ecological footprint and associated risks [1]. The practice of green chemistry is to introduce novel strategies for advancing sustainable practices with minimum bad environmental impact. Thus the frame work of philosophy of green chemistry laid the groundwork for effort to lessen the chemical industry's adverse pose on people and the planet [2].

Artificial intelligence (AI) is a result of revolutionary development in science and engineering [3]. In green chemistry, AI accelerates the development of sustainable materials and processes by optimizing resource utilisation. This achievement of AI' due to its ability to process large datasets, identify complex patterns and optimize systems [4]. , Now AI is being utilize as powerful tool for addressing critical sustainability challenges, such as reducing greenhouse gas emissions, minimizing waste, and decreasing dependence on non-renewable energy [5]– [6].One subfield of general chemistry that is devoted to reduce the use of potentially harmful substances in analytical procedures is known as "green analytical chemistry" (GAC). AI helps in reducing the use of harmful solvents and reagents and making use of direct analytics and miniaturised method approaches. The entire GC principle can be put into practice in a real laboratory setting by aglorithms [7]– [8]. Although significant progress has been made over the past three decades, many challenges remain, particularly in scaling up these practices for industrial applications and integrating them across diverse scientific disciplines [8].

Artificial Intelligence (AI) is defined by its ability to learn from data and apply that knowledge to make predictions or decisions [9]. In the context of green chemistry, AI has the potential to significantly accelerate research and innovation by automating routine tasks, analyzing large and complex datasets, and optimizing experimental conditions. For example, machine learning models have been successfully used to predict reaction yields and selectivity, thereby reducing the need for trial-and-error experimentation [9]. Similarly, AI-driven algorithms can determine optimal synthesis pathways for designing new catalysts or functional materials, improving both efficiency and sustainability [10, 11]. These applications demonstrate how AI is already contributing to green chemistry by minimizing resource consumption and enabling safer, cleaner laboratory practices [11]. – [12].

A powerful synergy can be achieved by integrating Artificial Intelligence (AI) with Green Chemistry (GC). This integration supports the transition from conventional fossil fuel based feedstocks to renewable biological resources by enhancing biocatalytic processes and expanding sustainable synthetic options through AI-powered predictive tools [13]– [14]. For instance, AI and machine learning models have been applied to optimize enzyme design and activity, enabling more efficient biocatalytic conversions of biomass into valuable chemicals [14]. Beyond synthesis, AI has become a revolutionary tool in academia and scientific research, improving laboratory workflows, accelerating large-scale data analysis, and fostering interdisciplinary collaboration [15]. However, the adoption of AI in green chemistry is not without challenges. A critical barrier remains the accessibility, quality, and reliability of experimental data, which are essential for training robust and generalizable AI models [16]. Addressing these issues is vital for maximizing the potential of AI-driven sustainable chemistry.

The convergence of Artificial Intelligence (AI), Data Science, and Green Chemistry represents a revolutionary frontier for environmental innovation. Intelligent computer systems have emerged as essential tools to address pressing global challenges such as climate change, resource depletion, and sustainable chemical manufacturing. Unlike traditional statistical approaches, data-driven AI possesses advanced analytical capabilities that enable the optimization of reaction conditions, the design of novel molecules, and the development of highly efficient chemical processes. For example, AI has been applied to accelerate the discovery of metal–organic frameworks for carbon capture and to design sustainable catalysts with improved selectivity [11]. This integration not only scales environmentally friendly technologies to new heights but also provides predictive capabilities and automation that were previously unattainable.

Positioning AI and Data Science at the core of Green Chemistry offers disruptive pathways toward long-term sustainability and environmental stewardship [17]– [18]. However, despite their remarkable ability to recognize complex patterns and generate predictions, many AI models lack interpretability, making it challenging for chemists to fully validate or understand the rationale behind AI-driven decisions [19]. Therefore, the effective use of AI in sustainable chemistry demands a multidisciplinary approach that brings together chemists, engineers, computer scientists, and data specialists. Developing user-friendly AI tools that can be seamlessly integrated into laboratory and industrial practices will be crucial to bridging these disciplinary divides [20]. With such collaborative efforts, AI has the potential to revolutionize industries ranging from pharmaceuticals to energy and materials, driving innovation while reducing environmental footprints, ultimately paving the way toward a greener and more sustainable future [21]– [22].

1.1 Overview of green and sustainable chemistry

The frame work and philosophy of green chemistry is based on twelve principles. These can be applied to any technique, process and product which consume or produce hazardous substances to develop more sustainable technique, process or product [23]. One example of a sustainable GC technique is the use of non-toxic solvents in place of potentially harmful ones. Perhaps the most famous area of GC is green analytical chemistry (GAC).The concept of Green Chemistry (GC) has evolved to a new level, now encompassing not only human health and the environment but also system-level considerations, application life cycles, and transdisciplinary activities [24]– [25]. Thirty years of scientific advancement have propelled framework of GC applied in chemistry's many branches forward. Industrialized nations now strive for a chemical industry that is sustainable and environmentally benign. Researchers and engineers anticipate that AI applications are poised to revolutionize all areas of chemistry and engineering aligned with green chemistry [26]. The chemical industry requires fresh insights into catalysis and novel processing methods to do away with conventional conversions, which frequently necessitate massive quantities of energy, release greenhouse gases (GHGs), deplete water supplies, and depend on rare and valuable metals.

Improving a chemical's life cycle requires researchers to focus on two main ideas: reducing risk and measuring risk events in a sustainable way [23].It is predicted that the global population will exceed 9 billion by 2055, making the conservation of food, fuel, fibre, and feed increasingly critical. To meet the demands of this future, we must

turn to renewable and sustainable alternatives [27]. Major challenge in Green Chemistry (GC) lies in its inherently interdisciplinary nature, which requires the integration of knowledge from diverse scientific and engineering domains to design safer processes and sustainable products. For instance, in nanotechnology, GC principles guide the synthesis of nanoparticles using eco-friendly methods such as plant extracts, biodegradable polymers, or microwave-assisted techniques, thereby avoiding toxic reducing agents. In biochemistry, GC contributes to the design of enzyme-catalyzed reactions and metabolic pathways that minimize hazardous by-products while maximizing efficiency. In the realm of biotechnology, the application of renewable feedstocks, biocatalysts, and microbial systems enables the production of biofuels, bioplastics, and fine chemicals with reduced environmental impact. Similarly, in materials science, GC promotes the development of sustainable polymers, green solvents, and recyclable composite materials [28].

Green Analytical Chemistry (GAC) emerged in year 2000, this is analytical chemistry aligned with principles of green chemistry avoid using hazardous solvents, reagents, preservatives, and other potentially dangerous substances [29]. Initially given top priority, GAC goes on to produce a plethora of laboratory based applications with the goal of mitigating chemical based adverse effects; eventually, it becomes one of the stimulants for analytical chemistry research. Several eco-friendly investigations have begun to develop in order to improve the precision of analytical outcomes [30]. Each chemical analysis procedure has several external parameters, including population, analyst, and research location (country), as well as multiple products and by-products, making the GAC multidimensional [31].

Research in conventional chemical sciences often leads to the development of materials that are not sustainable, being both resources intensive and environmentally detrimental over the long term [32]. To address these challenges, sustainable material design, supported by computer-assisted property prediction and simulation tools, represents the next major step in green research. In this context, a branch of Artificial Intelligence (AI) known as Machine Learning (ML) has gained significant attention in the sustainability movement over the past decade. ML offers innovative potential by improving process performance, enhancing material quality, and enabling accurate prediction of new sustainable materials, thereby accelerating innovation in engineering and chemical sciences [33].

1.2 Importance of AI in sustainability

Access to large, high-quality databases, tools is crucial for artificial intelligence technologies because software learning is make more competent them for development. If we wish to hasten advancements in the field of chemical sustainability, we should think about the two major obstacles that this argument brings to light. To begin, the computational and recording processes make it hard to get access to big, high-quality data sets that are pertinent to chemical research [33]. To build self-driven or AI-assisted laboratories, it is necessary to implement electronic lab books and results archiving systems in a systematic and standardized manner [34]. In order to ensure that AI development is fair and egalitarian, the second critical factor is open data, which is simply access to data [35]. Database of Reactions Concerning this matter, projects like The Open seek to create a centralized data repository and an open-access standard for organizing and sharing data

on organic reactions. The successful application of AI in sustainable research depends majorly on data availability and quality. Access to large, high-quality databases is crucial for AI technologies, since software learning and predictive modeling require well curated data sets. T o speed up progress in chemical sustainability, two main challenges need to be solved. First, computational and data recording barriers make it difficult to access large, reliable data sets relevant to chemical research [33]. To overcome this, the implementation of electronic lab notebooks (ELNs) and standardized results archiving systems is necessary for building self-driven or AI-assisted laboratories [34]. The second challenge is to make AI development fair and inclusive, which requires open data, meaning free and universal access to scientific information [35]. In this regard, initiatives such as the Open Reaction Database (ORD) aim to establish a centralized data repository and open access standards for organizing and sharing chemical reaction information. Such efforts provide the foundational infrastructure that enables AI and ML to effectively contribute to sustainable chemistry and materials innovation. (Table 1). Figure 1 shows importance of AI& DS in green and sustainable chemistry.

Table 1 Role of AI & DS to support applications of principles of GC and GAC

Principles of green chemistry (GC)	Principles of green analytical chemistry (GAC)	Role of AI and data science (DS)	References
Waste management	Sample treatments are reduced with the presence of direct analytics	AI enables predictive waste minimization. Machine learning models optimize reaction conditions to reduce by-products, while data-driven monitoring ensures efficient waste segregation.	[36]
Atom economy	Use minimum sample size and sample count	AI-driven retrosynthetic analysis identifies high atom economy pathways. Data science algorithms optimize reagent ratios and simulate product yields.	[37]
Less dangerous chemical synthesis	Presence of in-situ measurements	AI supports virtual screening for safer reagents. Real-time data analysis ensures controlled reactions, minimizing hazardous intermediates.	[38]
Safer chemical design and development	Reducing the usage of power and reagents through integration of analytical processes	Machine learning accelerates the molecular design of non-toxic compounds. AI optimizes energy consumption and resource utilization.	[39]
Safer solvents creation and auxiliaries	Creation of automated and miniaturized techniques	AI models predict sustainable solvent alternatives. Data science enables automated decision-making for solvent selection.	[40]
Improvement of efficiency by design	Neglecting the usage of derivatization	AI-driven optimization identifies the most efficient synthesis routes. Data science helps eliminate unnecessary analytical steps.	[41]
Increased usage of renewable feedstock	Reducing the production of analytical waste	AI predicts viable renewable feedstocks and bio-based alternatives. Data science supports life cycle assessments for sustainable resource use.	[42]
Reduce derivatives, catalysis, creation, degradation design	Multi-parameter techniques are introduced, power utilization minimized, renewable source-based reagents utilized	AI develops novel catalytic systems. Data science applies multi-objective optimization to minimize energy use and maximize renewable input.	[39]
Pollution prevention in real time	Neglecting the toxic reagents	AI-powered sensors and IoT provide real-time pollution monitoring. Data science predicts emission hotspots before they occur.	[43]
Accident prevention process	Highly concentrates on operator safety	AI-driven predictive maintenance reduces accident risks. Data science enhances safety analytics, ensuring safer laboratory and industrial operations.	[44]

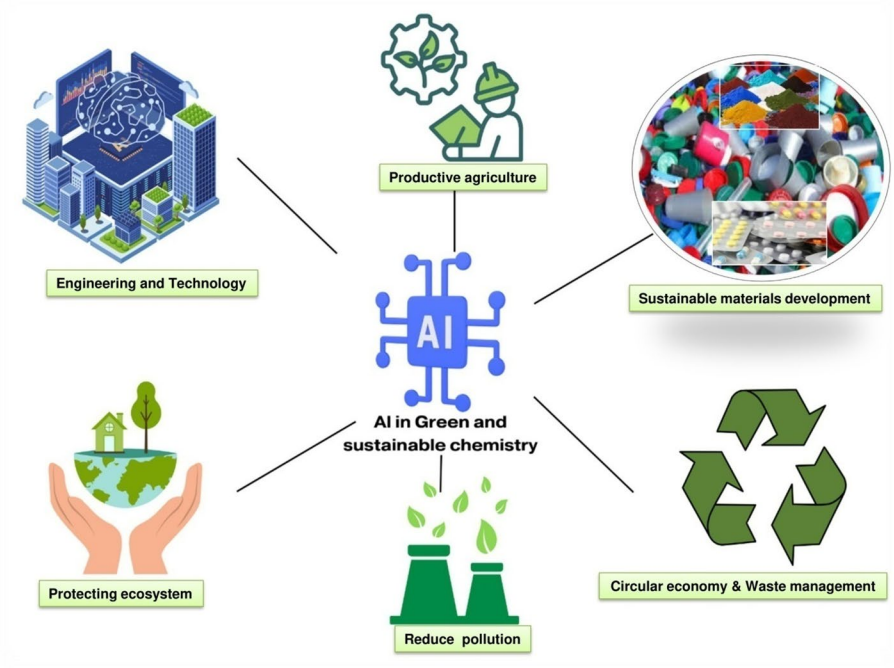

Fig. 1 Roles of AI in green and sustainable chemistry

Table 2 Areas of sustainable chemistry and AI application

Area of sustainable chemistry	Description	References
Organic synthesis	Use of green principles with AI in designing organic reactions	[49]
Catalyst development	AI helps design safe and eco-friendly catalysts	[50]
Analytical processes	Adoption of Green Analytical Chemistry (GAC) methods	[51]
Extraction processes	Development of safer and non-toxic extraction methods	[52]
Sample preparation	Greener approaches to sample collection, conditioning, and storage	[53]
Renewable products	Choosing materials based on sustainability features	[54]
Sustainable development	AI supports planning and education for green practices	[55]
Materials and inorganic substances	Greener methods for making inorganic and hybrid materials	[56]

1.3 Data science for green and sustainable chemistry

Data science builds on this foundation by providing the tools to analyze and use these datasets effectively. It allows patterns and insights to be extracted from complex experimental and environmental data. This supports reaction optimization, prediction of material properties and the discovery of sustainable alternatives. Methods such as cheminformatics, predictive modeling, multivariate analysis and statistical learning are now widely used to support decisions in reaction design, toxicity prediction and life cycle assessment. Data science also acts as a bridge between experimental work and computational research by enabling interpretable and transferable machine learning models. For example, machine learning models have been successfully applied to catalyst discovery, where algorithms can predict the activity and selectivity of catalysts before experiments are performed. This saves time and resources while reducing waste. Similarly, AI and data-driven tools are being used to design green solvents, where thousands of possible solvent candidates can be screened virtually to identify the most sustainable options for chemical reactions. With the growth of open-access databases and cloud

based platforms, data science makes research more reproducible and scalable. In this way, it is transforming traditional chemistry into a green sustainable and data-driven science [45].

2 Recent advances in a.i. For green and sustainable chemistry

Artificial intelligence (AI) is bringing new opportunities to develop greener chemical processes and materials. One important area is discovery of more efficient materials like green corrosion inhibitors, catalyst, where machine learning (ML) models trained on data from density functional theory (DFT) calculations help scientists quickly identify useful catalysts [45]. This method is called virtual high-throughput screening saves both time and resources by reducing the need for many experiments. AI is also helping in retrosynthesis, where large language models and natural language processing (NLP) predict the best reaction steps to make a desired compound [46]. Combined with automation and robotics, AI speeds up the discovery of new reactions and materials that support sustainable development [47]. By analyzing large experimental datasets, ML improves process optimization, product quality and environmental performance etc. in different area of sustainable chemistry Table 2. Another exciting direction is the use of AI with quantum mechanics to predict toxicity, which is vital for evaluating sustainability [48].

2.1 Machine learning models in chemical design

Machine learning (ML) helps scientists find patterns in chemical data and make predictions without doing endless experiments. This saves both time and resources. In chemistry, ML is now used to predict protein structures, design new molecules, and assess toxicity [57]. For green and sustainable chemistry, ML is very useful. It can design catalysts, predict material properties, and suggest greener reaction conditions [58]. A major problem is that chemical data is scattered in thousands of papers. Natural language processing (NLP) can help by reading these papers automatically and turning the text into useful data [59]. AI models work differently from old computer programs. Instead of following strict rules, they learn from past data and then make predictions on new cases [60, 61]. This is helpful in chemistry, where input–output links are often very complex. Large language models (LLMs) are a newer kind of AI. They are trained on huge amounts of text and can understand chemical writing, summarize results, and even suggest synthetic routes [62]. The idea of "foundation models," introduced in 2021, shows how these systems can be adapted for chemistry specific problems [63]. In practice, ML can replace or improve traditional equations. For example, instead of only using theory to estimate solubility, ML trained on compound structures can give faster and more accurate predictions [64]. With more computing power and better access to open data, ML is becoming an essential tool for sustainable chemistry. The range of applications of ML and LLMs in this field is summarized in Table 3.

2.1.1 Predictive toxicology and safer chemical design

AI is now helping chemists move from testing chemicals after they are made to designing safer ones from the start. Machine learning (ML) models, such as graph neural networks (GNNs), use large toxicology datasets like the U.S.(Environmental Protection Agencies)

Table 3 Applications of ML and LLMs in green and sustainable chemistry

Application in	Role of ML/LLMs	Practical application	References
Catalyst design	Predicts activity and stability	Screening eco-friendly catalysts	[58]
Reaction optimization	Suggests greener conditions	Reducing solvent or energy use	[59]
Material property prediction	Learns structure–property links	Forecasting solubility, toxicity, durability	[64]
Literature survey	Extracts data from papers	NLP finds reaction yields from text	[59]
Retrosynthesis	Plans synthetic routes	LLMs propose sustainable pathways	[62]
Education & knowledge sharing	Summarizes and explains results	AI-based tools for green chemistry learning	[63]

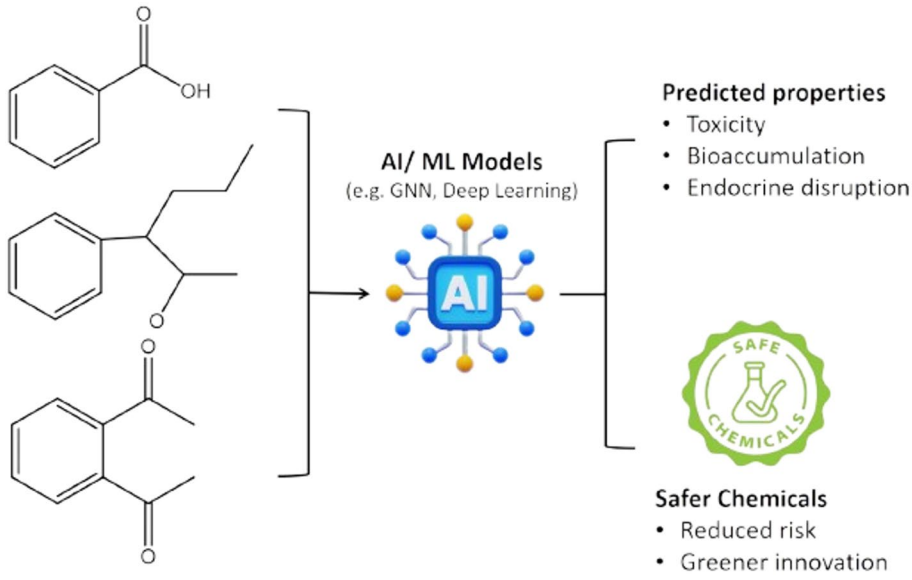

Fig. 2 AI in Predictive toxicology and safer chemical design

EPA's Tox Cast to predict properties such as toxicity, bioaccumulation, and hormone disruption [59, 62]. With deep learning, thousands of molecules can be screened on a computer before any synthesis takes place. This helps identify which candidates are more likely to be persistent, bioaccumulative, and toxic (PBT) Fig. 2 [64]. In this way chemists can avoid harmful compounds early and focus only on safer options. This approach is called safety by design and it represents a big change from the older method of risk assessment which came only after a chemical was produced. The main challenge is that these models depend on the data they are trained on. If a new compound is very different from existing data, the predictions can be less accurate. Expanding the coverage and diversity of toxicology datasets will make these models more reliable for future material design.

2.2 Intelligent reaction optimization and catalysis

Artificial intelligence (AI) is transforming the way reactions are optimized in green and sustainable chemistry. Traditionally, scientists relied on trial and error methods, which consumed a lot of time and resources. Now, AI can analyze large datasets that include solvents, catalysts, reactor setups and conditions like temperature and pressure to find the best reaction pathways [65, 66]. One of the most exciting advances is the rise of AI driven robotic platforms. These systems can autonomously adjust reaction conditions to

improve yield, reduce energy use and lower the environmental impact (measured by the E-factor). Instead of changing one variable at a time, methods like Bayesian optimization allow these platforms to explore many parameters at once such as catalyst loading, solvent choice, and residence time with far fewer experiments [3]. For catalyst development, machine learning (ML) models are now being used to find patterns that connect catalyst structures with their activity and selectivity. This makes it possible to design efficient catalysts based on earth-abundant materials, helping to replace scarce and expensive noble metals [4]. At the same time, AI models trained on density functional theory (DFT) and experimental results can predict how molecules adsorb on catalyst surfaces, which is crucial for understanding reaction mechanisms [67–72]. These predictions give insights into electronic structures, binding sites, and intermediates that form during catalysis [73, 74]. Such knowledge has already guided the discovery of new photocatalysts for water splitting, boosting solar-to-hydrogen energy conversion [75]. However, current systems have limitations. Many operate in a closed loop, meaning they can optimize reactions within a given setup but cannot easily propose new pathways or catalyst designs. To overcome this, researchers are working on integrating AI planning tools with robotic executors, to build fully autonomous systems for chemical discovery.

2.3 AI in green process development

Green chemistry focuses on developing cleaner and safer processes for chemical production, resource management, and pollution control. With the rise of artificial intelligence (AI), researchers are finding smarter ways to design, monitor, and optimize these processes for sustainability. One important area is advanced oxidation processes (AOPs) widely used for wastewater treatment. AI models can predict the efficiency of AOPs under different conditions, such as pH, oxidant concentration, and catalyst type, reducing the need for trial-and-error experimentation [76–78]. For instance, integrating machine learning with hydrodynamic cavitation or solar-assisted oxidation systems can improve pollutant degradation while lowering energy use. Another application is in adsorption and green adsorbent design. AI-driven models help in predicting adsorption kinetics and thermodynamics, guiding the selection of bio-based adsorbents such as plantain leaf biochar (PLB). Such approaches improve removal efficiency of dyes and industrial pollutants, while ensuring high regeneration rates and low waste generation [79–81]. Similarly, AI tools are accelerating the discovery and application of green solvents. Instead of relying on hazardous solvents for extraction and synthesis, predictive models now evaluate novel solvents on parameters like toxicity, solubility, and recyclability. This helps identify safer alternatives for biomass extraction and industrial processing [82]. AI is also playing a role in nanomaterial preparation for water treatment. For example, machine learning models can predict the size, morphology, and performance of iron oxide nanoparticles, which are commonly used for adsorption and pollutant degradation [83]. This reduces energy consumption and improves reproducibility in synthesis. Finally, AI contributes to sustainable energy-related processes. It supports catalyst design for renewable energy production, optimization of carbon capture methods, and evaluation of life cycle assessments to minimize environmental impact [84]. These advances highlight how AI integrates with green chemistry principles to make chemical processes cleaner, safer, and more efficient.

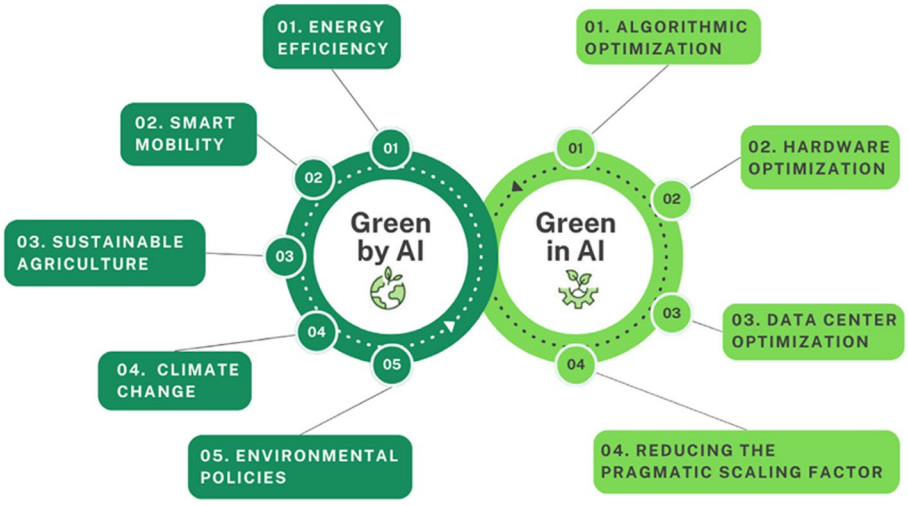

Fig. 3 Overview of Green by AI and Green in AI pathways highlighting energy efficiency, smart mobility, agriculture, climate change and AI hardware/algorithm optimization

3 Sustainable process design via AI

AI is playing an important role in redesigning chemical processes to make them more efficient, less polluting, and more sustainable [85, 86]. By combining experimental chemistry with data-driven tools, researchers are now able to optimise resources, reduce waste, and support the transition towards circular economy principles [87]. Two key areas where AI is making strong contributions are energy-efficient process design and waste minimization with recycling.

3.1 AI in energy-efficient process

One of the main challenges of a sustainable society is reducing energy demand in chemical production. AI-powered systems such as smart grids can optimise the balance between energy supply and demand, improving efficiency compared to traditional one-way grids [88]. These systems rely on real-time data and predictive algorithms to manage renewable energy sources like solar and wind, which are variable and hard to forecast. The broader role of AI in sustainability can be understood through the concepts of "Green by AI" (using AI to make processes more sustainable) and "Green in AI" (making AI itself more sustainable), as shown in Fig. 3.

Machine learning models, including deep neural networks and hybrid architectures, are now being applied to renewable energy forecasting, helping integrate clean energy into chemical manufacturing [89, 90]. Similarly, in industrial settings, AI can reduce energy consumption by optimising heating, ventilation, air conditioning (HVAC) and lighting systems in chemical plants and laboratories. These optimisations directly reduce carbon emissions and operational costs, showing how AI contributes both economically and environmentally [90].

3.2 Waste minimization and recycling with AI

Waste management remains a core challenge in chemical industries. Traditional methods to predict and track pollution are limited due to the complex and dynamic nature of environmental data. AI is now being used to process information from satellites, sensors, and monitoring stations to, trace sources, and model the behaviour of contaminants in

air and water systems [91, 92]. In recycling, it can improve the separation and recovery of valuable materials from industrial and municipal waste. For instance, predictive models can identify optimal conditions for adsorption in wastewater treatment or forecast how organic compounds interact with soil, enabling smarter design of remediation strategies [93–95]. AI-driven approaches are also being tested for, where real-time monitoring combined with predictive models helps optimise treatment processes while lowering chemical and energy inputs [92]. These applications show how AI can support the reduce, reuse and recycle model central to green and sustainable chemistry.

4 Environmental impact assessment through AI

Artificial Intelligence (AI) is increasingly being used to understand and reduce the environmental impacts of chemicals and industrial processes. Instead of relying only on experiments or lengthy assessments, AI allows us to predict hazards, estimate environmental footprints and even design safer alternatives before large-scale production begins. This makes AI a powerful tool for sustainable chemistry [96–98].

4.1 Predictive models for toxicity and environmental fate

Traditionally, toxicity testing has depended on animal experiments, which are costly, time-consuming, and ethically challenging. With AI, researchers can now build models that predict how a chemical will behave in the body and in the environment [99–101]. For example, machine learning systems trained on large toxicology databases can estimate whether a substance is likely to be persistent, bioaccumulative, or toxic. This helps scientists avoid harmful molecules early in the design stage. These tools are also valuable for predicting how chemicals spread in soil, water, or air. By combining data from laboratory studies with computational models, AI can give early warnings about substances that may cause long-term ecological problems [102–104]. This shift from "test and react" to "predict and prevent" represents a major step toward safer chemical design.

4.2 AI tools for reducing carbon footprint

Beyond only lowering their impact on the environment, there are numerous advantages to making algorithms more efficient. Designing optimisation techniques that reduce computational resource requirements, thereby minimising energy usage, is one of the most productive tactics in green algorithm development. Reduced memory footprint and computational complexity of training models are active areas of research in areas such as sparse training approaches, quantisation techniques, energy-aware pruning, and low-precision arithmetic operations [105]. While some studies have suggested more efficient methods, others have detailed scalable and sustainable methodologies for condition-based maintenance of wheel out-of-roundness using locality-sensitive hashing to analyse large time series datasets, and yet others have developed new and efficient methods for feature selection in domain adaptation using mutual information maximisation (MIM) instead of the commonly used evolutionary algorithms [106]. Energy savings can be achieved by selecting more computationally efficient hardware. For example, certain graphics processing units (GPUs) have far higher efficiency in terms of floating point operations per second (FLOPS) per watt of power utilisation than other GPUs. Another type of specialised hardware accelerator is the tensor processing unit (TPU), which is

designed to handle machine learning tasks and may be configured to employ ML models that are optimised for that particular hardware [107].

Importantly, increasing the number of cores to 15 improves execution times and reduces GHG emissions for a particular workload, suggesting that parallelisation is a natural approach to reducing algorithm training time [108]. The authors' tests show that tiny increases in runtime lead to big emissions, but when the reduction in execution time is smaller than the proportional increase in the number of cores, emission levels get worse.

Lastly, one important strategy in this context is edge computing. The idea is to do computation close to where the data is collected or used, rather than sending it to a data centre or the cloud. This helps with battery life and processing power limitations of IoT devices [109]. How efficient a data centre is and how carbon intensive its location is both contribute to its carbon impact. Because there is a wide range of values among nations, the latter is perhaps the most crucial component when considering overall carbon footprint. For example, in Norway and Switzerland, it is less than 20 gCO2e kWh − 1, whereas in Australia, South Africa, and some US states, it exceeds 800 gCO2e kWh − 1.Researchers have devised algorithms and frameworks to optimise data centre utilisation, with the goal of reducing energy consumption through dynamic server load management, cooling system adjustments, and optimal resource allocation [110]. Without a doubt, the simplest approach to decrease power consumption is to limit the frequency of algorithm runs, particularly computationally intensive ones [111].

4.3 Life cycle assessment (LCA) and supply chain optimization

Every chemical product has an environmental story from the resources used to make it, to how it is transported and finally how it is disposed of. This is what Life Cycle Assessment (LCA) tries to capture. Traditionally, gathering such data has been slow and labor-intensive, but AI can make the process faster. Natural language processing, for example, can scan thousands of patents and papers to build large life cycle databases in a fraction of the time. Beyond LCA, AI is also changing how supply chains are managed. In chemical industries, raw materials often come from many sources and involve complex logistics. AI-based optimisation models can help design supply chains that are not only cost-effective but also reduce emissions, minimise waste, and ensure responsible use of natural resources [111]. This is especially important for sustainable chemistry, where both production methods and distribution systems need to align with environmental goals.

With these approaches show how AI is reshaping environmental assessment from predicting toxicity and lowering carbon footprints to improving life cycle analysis and supply chains. By combining these tools, sustainable chemistry can move beyond reactive measures and design processes that are safer, cleaner and more efficient from the start.

5 Challenges and limitations

5.1 Data availability and quality

AI in chemistry depends on data, but often the datasets are limited or inconsistent. Many reaction datasets are small, incomplete, or reported in formats that are not standardized. In practice, even moving computing to efficient cloud platforms can reduce emissions and energy use by 25–50% compared to local systems, yet this efficiency does

not solve the fundamental issue of poor data quality [112]. Real-world data centres are often smaller and older, which prevents them from fully exploiting advanced cooling and energy-saving technologies [113, 114]. Therefore, alongside infrastructure improvements, it is essential to develop FAIR (Findable, Accessible, Interoperable, Reusable) datasets to overcome the current bottleneck in robust model training and generalization [115].

5.2 Interpretability of AI models

Machine learning can predict reaction outcomes or label a chemical as "green," but the reasoning is not always clear. This "black box" nature limits trust in AI models, especially for scientists who need mechanistic understanding. Even though advanced hardware like TPUs and neuromorphic chips offer major efficiency gains and reduce the carbon footprint of model training, they do not address the lack of interpretability [115]. The emerging field of explainable AI (XAI) in chemistry is therefore crucial to connect predictions with chemical intuition and scientific insight.

5.3 Scalability and industrial adoption

Scaling AI tools from laboratory settings to industry is not straightforward. Chemical plants and industrial processes operate under strict regulatory, safety, and cost constraints. Even if AI suggests greener routes, adoption is often slowed by concerns about economic feasibility and process stability. Cloud providers, by increasingly relying on renewable energy sources such as wind, solar, and hydro, can reduce carbon emissions by a factor of 5–10 compared to individual on-premises systems [115]. Still, industrial integration will require collaboration between chemists, engineers, and policymakers to ensure responsible and sustainable deployment. Key challenges that cut across these limitations include:

(i) *Data gaps and inconsistency* Chemistry does not yet have large, standardized datasets like those available in other AI-heavy fields [112–114]. Much of the existing data is either incomplete or reported in inconsistent formats, making it hard to build reliable AI models.
(ii) *Black-box predictions* Many machine learning models make accurate predictions but cannot explain the reasoning behind them [115]. This lack of transparency limits scientific trust and slows the discovery of new chemical insights.
(iii) *Energy footprint of AI* Training and running large AI models requires massive amounts of energy [112]. If not managed carefully, this energy use can offset the very sustainability goals that AI aims to achieve.
(iv) *Interdisciplinary barriers* Chemists, computer scientists, and engineers often face communication gaps, which slows down real progress [115].

6 Emerging opportunities and future directions of AI in green chemistry

AI is opening new frontiers in sustainable chemistry by accelerating discovery, improving efficiency, and reducing environmental impact. By analyzing large datasets, AI can quickly identify promising materials, optimize chemical reactions, and suggest safer

formulations. This reduces time, energy, and waste compared to traditional trial-and-error approaches [92, 101]. One important area is biodegradable materials. AI models can predict which polymers are more likely to break down safely, helping scientists design alternatives to conventional plastics [92]. In renewable energy, AI is guiding the development of efficient solar cells and biofuels, cutting emissions and improving performance [101]. In pharmaceuticals, AI helps find greener synthesis routes, lowering hazardous waste and enabling better solvent recovery [101].Beyond these fields, several exciting directions are emerging:

(a) *Generative AI for molecular discovery* New models such as VAEs and GANs can design entirely new molecules that are biodegradable, safe, and high-performing, moving beyond simple prediction [116].
(b) *Circular chemistry and plastic recycling* AI and robotics are improving plastic waste management by sorting polymers and discovering enzymes and processes for plastic depolymerization [117].
(c) *Robotic laboratories* Self-driving labs, guided by AI, can plan and perform experiments automatically, creating a feedback loop that speeds up sustainable discovery [118].
(d) *Hybrid AI-knowledge systems* Combining data-driven ML with chemical knowledge graphs makes AI reasoning more transparent and trustworthy, addressing the "black-box" challenge [119].

Together, these advances show that AI is not just a tool for optimization but a driver of innovation. As these technologies mature, they will play a central role in building greener, safer, and more sustainable chemical industries.

7 Future perspectives

Artificial intelligence is expected to play a growing role in advancing green chemistry in the coming years. Future applications may focus on tailoring sustainable chemical solutions to local contexts by integrating regional climate, resources, and policy needs. AI can also accelerate cross-disciplinary collaboration, enabling chemists, engineers, and computer scientists to jointly design innovative catalysts, biodegradable polymers, and renewable energy systems. In addition, AI-driven life cycle assessment and supply chain optimization will help industries minimize waste, energy use, and environmental footprint. Public engagement through AI-based educational tools and open platforms may further promote awareness and adoption of sustainable practices. Finally, linking AI with autonomous laboratories and Industry updated technologies will create a more circular and resource-efficient chemical sector, where discovery, production, and recycling are seamlessly connected.

8 Conclusion

This all-encompassing analysis concludes that GC is critical to the development of a sustainable society. This investigation into GC's foundational principles and processes illuminates the intricacies of green solvents, sustainable synthesis, catalysis, and reaction conditions. There is a school of thought among academics that claims AI and ML have

tremendous promise for revolutionising Green Chemistry and bringing about a more sustainable future. The integration of artificial intelligence and data science into green and sustainable chemistry holds potential for building a cleaner, more resilient future. Advancing this synergy through collaborative frameworks and open data ecosystems is key to achieving impactful, long-termsustainability.AI has the potential to significantly advance efforts to create a cleaner, greener society by speeding up the creation of environmentally friendly materials, improving chemical processes, and facilitating better cooperation amongst experts. Never lose sight of the fact that AI is merely a component, though. A better, more sustainable future for future generations is within our reach if we band together and harness the potential of AI.

Author contributions
Shailendra Yadav conceptualized the study, coordinated the study framework, and first draft the manuscript. Harish Chandra and Kanha Singh Tiwari contributed significantly to literature collection. Manoj Kumar Bharty assisted in refining the discussion and provided valuable insights into data science applications. Rishikesh Chandravanshi and Dheeraj Singh Chauhan supported the technical analysis, and contributed to editing the manuscript. Mumtaz Ahmad Quraishi supervised the manuscript and critically reviewed the content for scientific accuracy, and approved the final version for submission. All authors read and approved the final manuscript.

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